

# FondoMexicoAlfa (FMIA)

Systematic Long-Only Equity & FIBRA | Quantitative Research Report

**-0.18**

Sharpe Ratio

**5.25%**

Annualized Return

**14.56%**

Annualized Vol

**-39.83%**

Max Drawdown

**-2.11%**

CVaR 95% (daily)

**3.48%**

Avg Turnover/Rebalance

Headline metrics from live run: source=bloomberg, model=elasticnet, as\_of=2026-03-31.

Walk-forward validated \* Bloomberg data \* Mexican equities & FIBRAs (BMV) \* Monthly rebalancing \* 10 bp transaction cost

Report date: 2026-05-19

## Strategy

|                    |                                                          |
|--------------------|----------------------------------------------------------|
| <b>Universe:</b>   | ~26 Mexican equities + FIBRAs listed on BMV              |
| <b>Signal:</b>     | LightGBM cross-sectional return forecast (RandomizedSear |
| <b>Baseline:</b>   | ElasticNetCV - interchangeable via config flag           |
| <b>Optimizer:</b>  | Mean-variance (SLSQP) with market-impact penalty         |
| <b>Rebalance:</b>  | Monthly, end-of-month, walk-forward OOS                  |
| <b>Validation:</b> | 108 rebalances, 2017-2026                                |

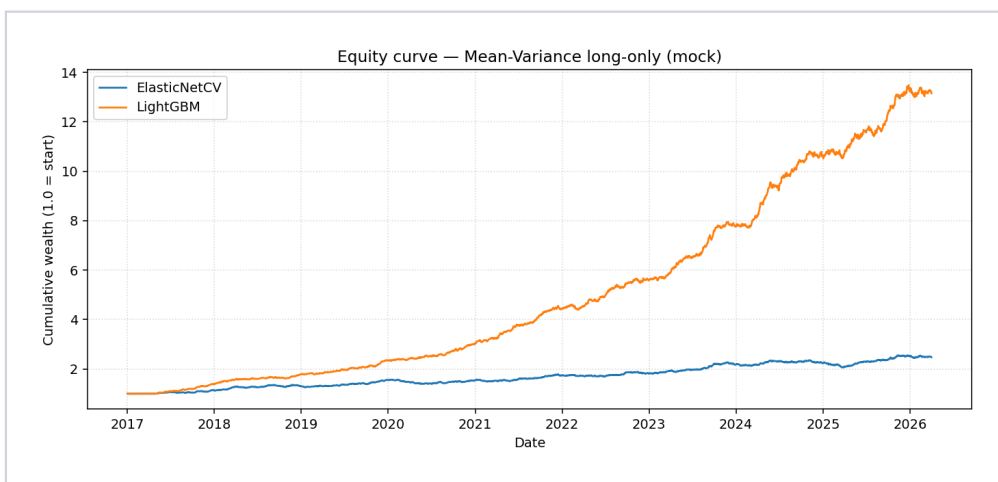
## Risk Controls & Attribution

|                      |                                                          |
|----------------------|----------------------------------------------------------|
| <b>Costs:</b>        | 10 bp per side applied each rebalance                    |
| <b>Regimes:</b>      | Banxico rate (TIGHTENING/EASING/NEUTRAL) x IPC vol stres |
| <b>Attribution:</b>  | Per-rebalance SHAP values via TreeExplainer              |
| <b>Safety valve:</b> | Revert to ElasticNet if top-5 SHAP stability < 0.30 for  |
| <b>Data:</b>         | Bloomberg (production) / mock data (research)            |
| <b>Framework:</b>    | CNBV position limits, LFI structure                      |

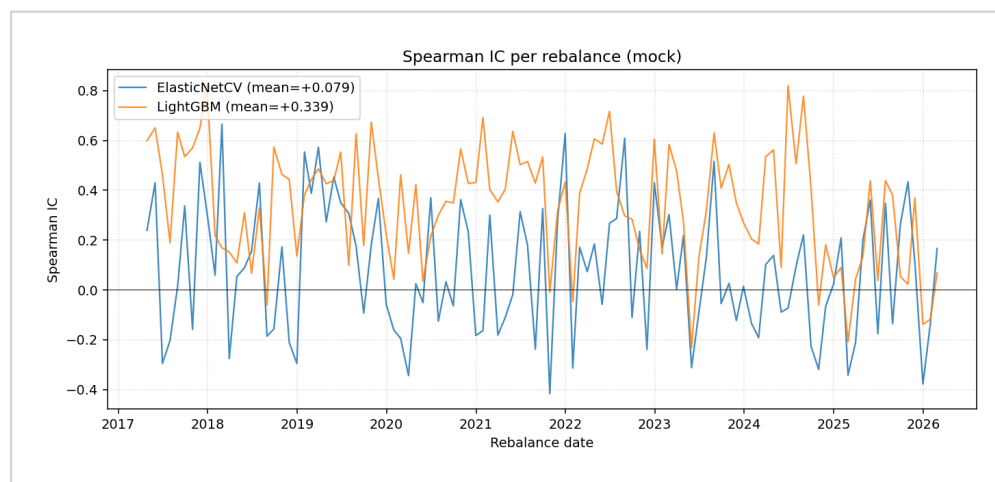
Headline metrics from live run: source=bloomberg, model=elasticnet, as\_of=2026-03-31.

## 01 MODEL PERFORMANCE - LightGBM vs ElasticNetCV (mock data - pipeline validation only)

| Metric                   | ElasticNetCV | LightGBM | D (lgbm - elastic) |
|--------------------------|--------------|----------|--------------------|
| IC mean (Spearman)       | +0.0794      | +0.3389  | +0.2595            |
| IC std                   | 0.2597       | 0.2335   | -0.0263            |
| ICIR                     | +0.306       | +1.452   | +1.1460            |
| Hit rate (daily)         | 0.516        | 0.573    | +0.0564            |
| Annualized return        | +0.0995      | +0.3089  | +0.2094            |
| Annualized vol           | 0.0763       | 0.0757   | -0.0006            |
| Sharpe                   | +0.981       | +3.292   | +2.3100            |
| Sortino                  | +1.025       | +3.626   | +2.6012            |
| Max drawdown             | -0.127       | -0.046   | +0.0806            |
| CVaR 95% (daily)         | -0.0097      | -0.0091  | +0.0006            |
| Turnover (per rebalance) | 0.0624       | 0.2684   | +0.2060            |
| Forecast wall time (s)   | 13.0         | 573.0    | +560.0181          |



Cumulative equity curves (108 rebalances)



Rolling information coefficient (IC) time series

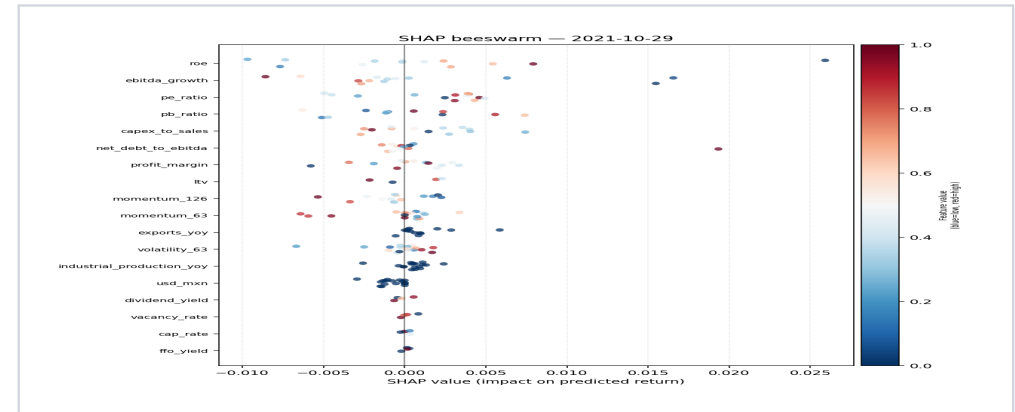
## 02 FEATURE ATTRIBUTION (SHAP) - LightGBM Walk-forward

### Top-10 Features by Mean |SHAP|

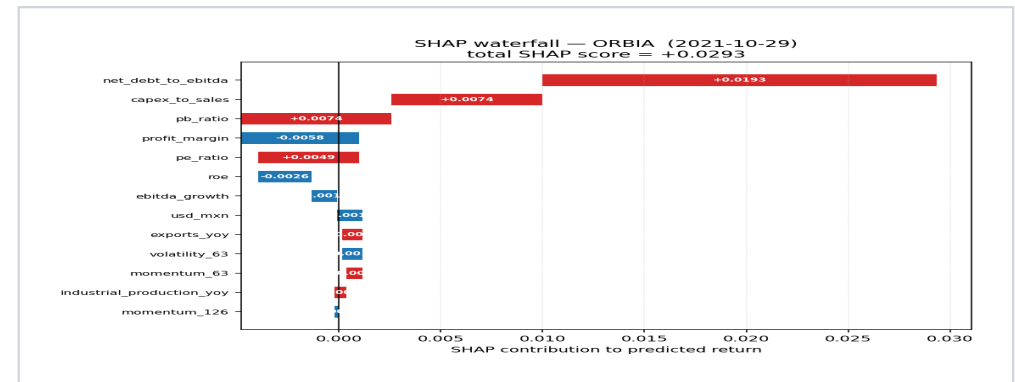
| #  | Feature        | Mean  SHAP | Std     |
|----|----------------|------------|---------|
| 1  | ltv            | 0.00350    | 0.00363 |
| 2  | ffo_yield      | 0.00279    | 0.00360 |
| 3  | cap_rate       | 0.00221    | 0.00235 |
| 4  | pe_ratio       | 0.00196    | 0.00233 |
| 5  | dividend_yield | 0.00182    | 0.00214 |
| 6  | momentum_63    | 0.00172    | 0.00249 |
| 7  | roe            | 0.00154    | 0.00183 |
| 8  | ebitda_growth  | 0.00146    | 0.00188 |
| 9  | profit_margin  | 0.00145    | 0.00220 |
| 10 | capex_to_sales | 0.00136    | 0.00219 |

### Feature Stability (Spearman rank-correlation)

| K      | Pairs | Mean r | Std r |
|--------|-------|--------|-------|
| ---    | ---   | ---    | ---   |
| top-5  | 107   | 0.440  | 0.421 |
| top-10 | 107   | 0.428  | 0.329 |
| all    | 107   | 0.455  | 0.292 |



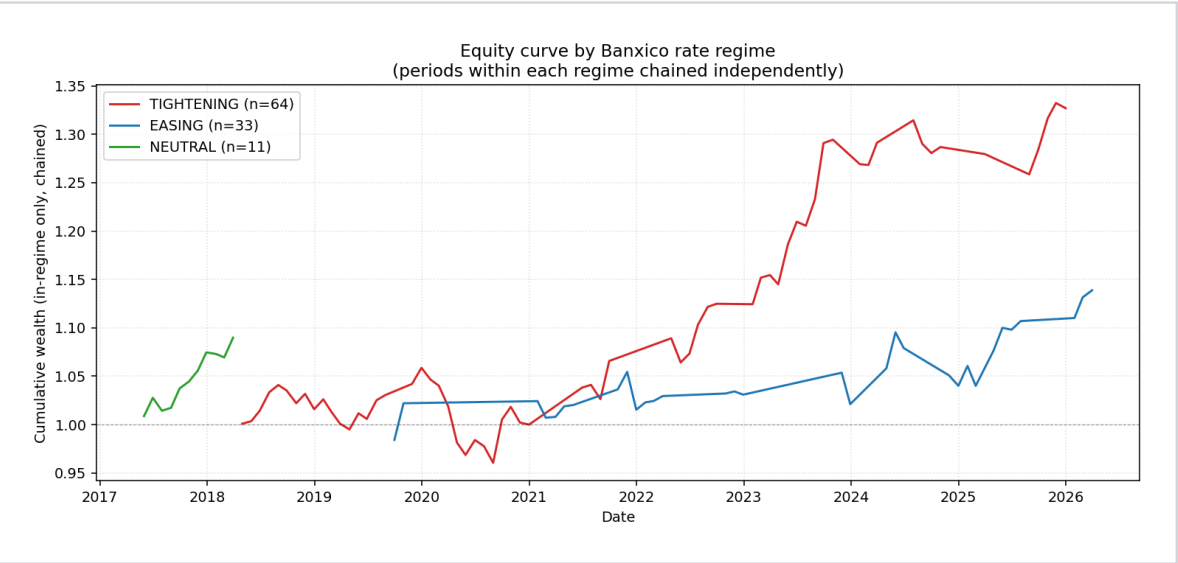
Beeswarm - Midpoint Rebalance (2021-10-29)



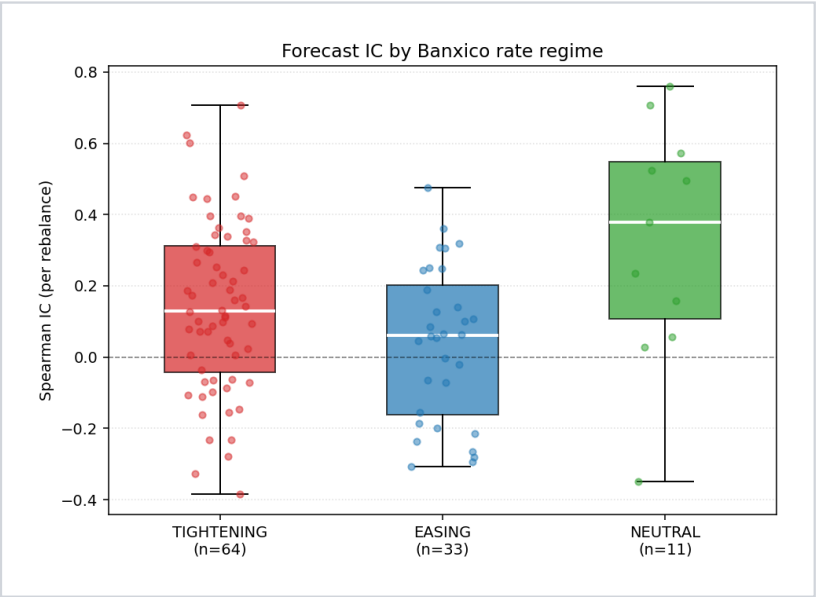
Waterfall - Highest-Predicted Stock

03 PERFORMANCE BY MACRO REGIME (mock data - pipeline validation only)

| Rate regime | Stress | N  | IC mean | Sharpe | SHAP top-5 stab. |
|-------------|--------|----|---------|--------|------------------|
| EASING      | ALL    | 33 | +0.039  | +1.27  | 0.566            |
| EASING      | CALM   | 23 | +0.021  | +1.74  | -                |
| EASING      | STRESS | 10 | +0.115  | +1.58  | -                |
| NEUTRAL     | ALL    | 11 | +0.280  | +2.27  | 0.386            |
| NEUTRAL     | CALM   | 11 | +0.280  | +2.27  | -                |
| TIGHTENING  | ALL    | 64 | +0.140  | +1.01  | 0.407            |
| TIGHTENING  | CALM   | 47 | +0.195  | +1.07  | -                |
| TIGHTENING  | STRESS | 17 | -0.035  | +1.66  | -                |



Regime-conditioned cumulative equity curves



IC distribution by macro regime

04 RISK PROFILE & METHODOLOGY

ElasticNet Regulated - Bloomberg OOS (source of truth)

| Metric                 | Value   |
|------------------------|---------|
| Annualized return      | 5.25%   |
| Annualized vol         | 14.56%  |
| Sharpe ratio           | -0.18   |
| Max drawdown           | -39.83% |
| CVaR 95% (daily)       | -2.11%  |
| Avg turnover/rebalance | 3.48%   |

Headline metrics from live run: source=bloomberg, model=elasticnet, as\_of=2026-03-31.

LightGBM Risk Metrics (Walk-forward OOS - mock data)

| Metric                 | Value   |
|------------------------|---------|
| Annualized return      | +0.3089 |
| Annualized vol         | 0.0757  |
| Sharpe ratio           | +3.292  |
| Sortino ratio          | +3.626  |
| Max drawdown           | -0.046  |
| CVaR 95% (daily)       | -0.0091 |
| Avg turnover/rebalance | 0.2684  |

Methodology

|                  |                                                                      |
|------------------|----------------------------------------------------------------------|
| Universe:        | Mexican equities + FIBRAs (BMV), ~26 instruments                     |
| Signal:          | LightGBM cross-sectional return forecast (RandomizedSearchCV + TimeS |
| Baseline signal: | ElasticNetCV (interchangeable via config flag)                       |
| Optimizer:       | Mean-variance (SLSQP) with market-impact penalty                     |
| Rebalance:       | Monthly (end-of-month), walk-forward OOS                             |
| Validation:      | 108 rebalances, 2017-2026                                            |
| Costs:           | 10 bp per side applied each rebalance                                |
| Regimes:         | Banxico rate (TIGHTENING/EASING/NEUTRAL) x IPC vol stress            |
| Attribution:     | Per-rebalance SHAP via TreeExplainer (shap 0.51)                     |
| Data:            | Bloomberg (production) * mock data (research)                        |
| Framework:       | CNBV position limits, LFI structure                                  |

DISCLAIMER: For research purposes only. Past performance does not guarantee future results. All results based on walk-forward out-of-sample simulation. Mock data used where Bloomberg data unavailable. This document is confidential and intended solely for internal research use.